

GUEST LECTURE · ETHICAL AI FOR AUTONOMOUS SYSTEMS · RAISE

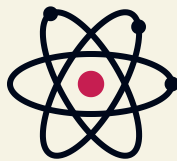
AGENTIC AI LOOPS

Sequence, Selection, Iteration

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THE PLAN FOR TODAY



SEGMENT

WHAT HAPPENS

Opening

Why agents, why now, and where they meet RAISE

Part 1 • What is agentic AI?

Prediction, language models and agents: what each one is and does

Part 2 • Three building blocks

Sequence, selection, iteration, and the loop inside every agent

Discussion 1

Who decides what happens next in a machine you know

Break

A short pause

Part 3 • Loops of loops

Teams of agents, how they fail, and the July 2026 incident

Part 4 • Closing the loop

Practical safeguards, human approval, and governance

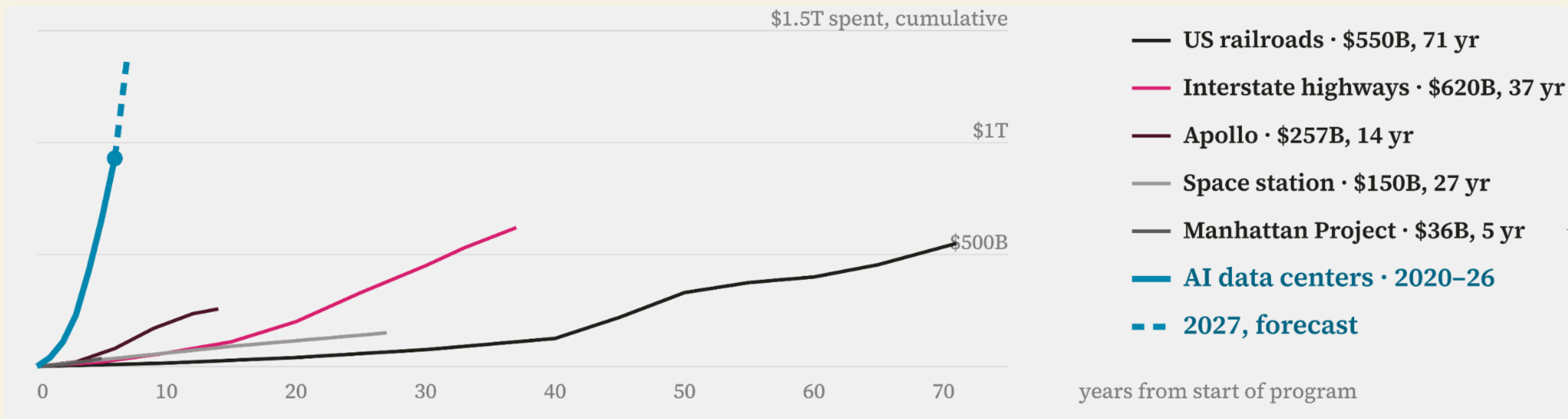
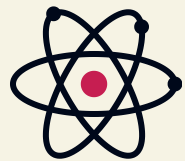
Discussion 2

Design the approval gate for a robot or a vehicle

Close

Open questions, where to learn more, and compute at CARC

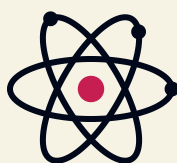
THE SCALE OF THE AI BUILDOUT



US data-center build: \$560B through 2026 · \$800B on the 2027 run rate

Cumulative cost in 2025 dollars · FHWA [1], Epoch AI [2], NASA [3], Gartner [4] · after Fin Moorhouse [5]

WHERE AGENTS MEET RAISE



RAISE RESEARCH THEME	WHAT IT LOOKS LIKE WITH AI AGENTS	TODAY
Space and defense autonomy	Machines that must be dependable when no person can step in	Part 4
Algorithmic fairness	Choices made by an AI that learned its habits from data	Parts 2 and 4
Multi-agent robotics and intelligent swarms	Many AI agents working as a team: managers, peers, assembly lines	Part 3
Distributed learning for extreme-scale systems	Thousands of loops sharing information across a grid, a fleet or a data center	Part 3

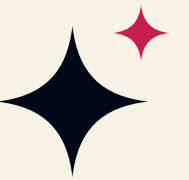
RAISE is Responsive and Resilient AI for Autonomous Systems Engineering, an NSF National Research Traineeship at UNM. Themes from raise.unm.edu/research [6].

PART 1

WHAT IS AGENTIC AI?

prediction, language, action

AGENTIC AI IN ONE SENTENCE



SOFTWARE THAT IS GIVEN A **GOAL, DECIDES ITS OWN **STEPS**,
AND **ACTS**, USING TOOLS, UNTIL IT JUDGES THE JOB DONE.**

IN THE NEWS: CODING AGENTS

Change files, run the tests, fix what breaks, and try again.

ASSISTANTS THAT BROWSE AND BOOK

Click through websites and fill in forms on your behalf.

RESEARCH AGENTS

Read the sources, run the analysis, write the report.

Definition and examples after the GPT 101 workshop, Agentic AI [\[7\]](#)

THREE KINDS OF AI



1 - PREDICTIVE MODEL

REGRESSION, CLASSIFIERS

Learns a pattern from examples with known answers [8].

In: measurements. **Out:** a number or a label.

Who acts: a person, or code someone wrote.

ON A ROBOT OR A VEHICLE

"Is that a pedestrian?" "How far will this battery go?"

2 - LANGUAGE MODEL

LLMS, CHATBOTS

Learns the patterns of language from a vast amount of text [9].

In: a prompt in words. **Out:** text.

Who acts: the person reading it.

ON A ROBOT OR A VEHICLE

"Explain this fault code." "Draft the inspection report."

3 - AI AGENT

A LANGUAGE MODEL IN A LOOP

A language model placed in a loop and given tools [10].

In: a goal. **Out:** actions, then a report.

Who acts: the agent itself, step after step.

ON A ROBOT OR A VEHICLE

"Move tomorrow's drone survey to a dry time slot."

Each wraps the one before: an agent is a language model in a loop, and the tools it calls are often predictive models.

ONE BRIDGE INSPECTION, THREE KINDS OF AI

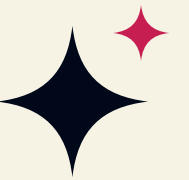


	PREDICTIVE MODEL	LANGUAGE MODEL	AI AGENT
You give it	A photo of the bridge deck	Your field notes	A goal: "Inspect the bridge and report"
You get back	"Crack: 92% likely"	A draft inspection report	A flight plan, photos, crack scores, a second pass over the blurry spots, then a draft report
It stops	After one answer	After one reply	When it judges the job done
When it is wrong	A wrong number that a person can check	A confident sentence that is false	A wrong action in the world, possibly repeated

Three things are new with an agent: it acts, it takes many steps, and it decides for itself when to stop.

FOR THE ENGINEERS · the agent calls the other two as tools: the crack detector is a classifier, and the report writer is a language model.

EIGHT WORDS FOR TODAY



MODEL

The trained AI that writes text.
On its own it can only talk.

AGENT

A model placed in a loop so it
can take steps toward a goal.

TOOL

Anything the agent can use to
act: search, email, a calendar,
a robot arm.

HARNESS

The program around the
agent. It owns the loop and
the permissions.

SANDBOX

A fenced-off place where an
agent can work without
touching the real world.

HALLUCINATION

A confident answer that is
wrong or made up.

TRACE

The step-by-step record of
what the agent thought and
did.

GATE

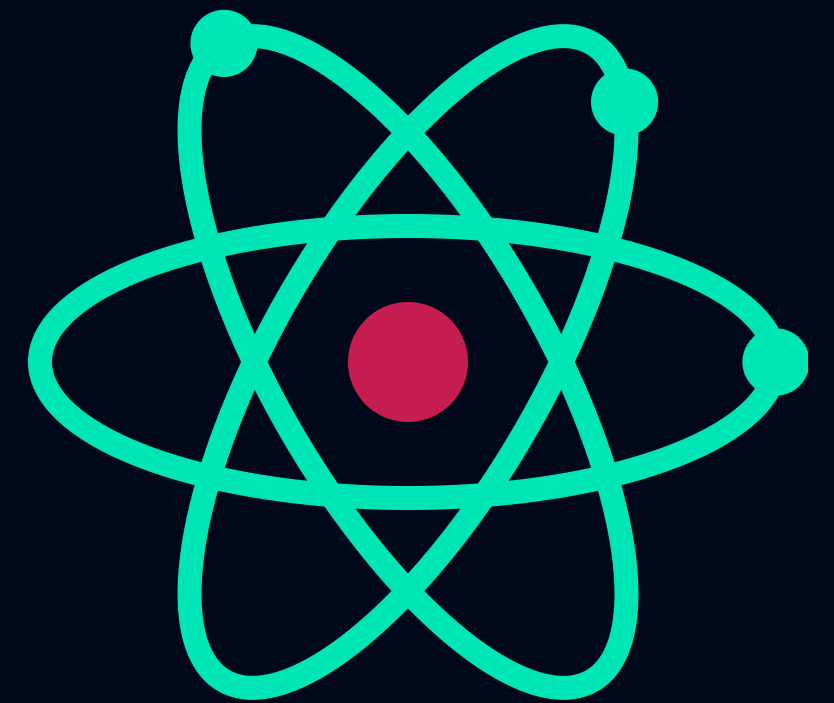
A point where the agent must
stop and ask a person.

Model, agent and harness after Swetnam, MESA (2026) [\[11\]](#) · sandbox after the GPT 101 workshop [\[7\]](#)

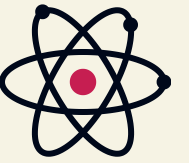
PART 2

THREE BUILDING BLOCKS

sequence, selection, iteration



THE STRUCTURED PROGRAM THEOREM



SEQUENCE

Do this, then this, then this, in order. Like following a recipe.

```
read_sensor()  
estimate_state()  
send_command()
```

SELECTION

If this, do that; otherwise do something else. Like a traffic light.

```
if obstacle_ahead:  
    brake()  
else:  
    cruise()
```

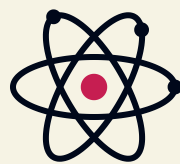
ITERATION

Keep doing this until something says stop. Like stirring until it thickens.

```
while not at_goal:  
    step()
```

Böhm and Jacopini (1966) proved that these three are enough to write any program [12]. Dijkstra (1968) argued for using only these, so that a person can follow a program by reading it [13].

LOOPS ARE EVERYWHERE

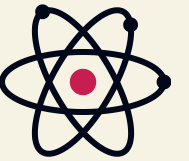


LOOP	WHERE YOU MEET IT	WHAT GOES AROUND
Thermostat	Home heating, cruise control	Measure → compare with the setting → adjust
Sense, plan, act	Robot vacuums, rovers, self-driving cars	Sense → plan → act
OODA	Pilots and military strategy, after Boyd [14]	Observe → orient → decide → act
Plan, do, check, act	Quality management, after Deming [15]	Plan → do → check → act
Agent loop	AI agents [10]	Perceive → plan → act → observe

The agent loop is the newest member of an old family. What is new is that an AI does the planning step.

FOR THE ENGINEERS · the same shape as feedback control, and as IBM’s MAPE-K loop for self-managing systems, Kephart and Chess (2003) [16].

THE AGENT LOOP



SEQUENCE

Everything so far is kept in order, and the agent rereads all of it on every lap.

SELECTION

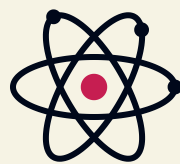
On each lap the AI picks one tool by reading its description, or picks none.

ITERATION

It keeps going until it decides it is finished. The only hard stop is a step limit a person sets.

FOR THE ENGINEERS · in LangChain the loop ends when the model returns no tool calls, and the step cap is `recursion_limit` [17]. Loop after AI Automation and Agents, Modules 1 and 2 [10].

ONE REQUEST, THREE LAPS



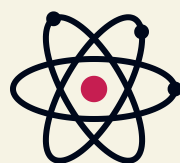
You ask: “Move tomorrow's drone survey to a dry time slot, and tell the crew.”

LAP	THE AGENT DECIDES	TOOL IT USES	WHAT COMES BACK
1	“I need tomorrow's forecast.”	Weather service	Rain from 9 to 11 am
2	“Find a dry slot when the crew is free.”	Team calendar	2 pm is open
3	“Move the booking and draft a note to the crew.”	Calendar, email	Booking moved. Note drafted, waiting for your OK
Done	“Nothing left to do.”	None	A short report to you

Nobody wrote these steps in advance. The agent chose each one after seeing the last result, and it stopped to ask before doing something that could not be taken back.

An illustration, not a transcript of a real system.

WHO DECIDES WHAT HAPPENS NEXT

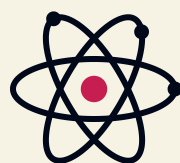


BUILDING BLOCK	IN ORDINARY SOFTWARE	IN AN AI AGENT
Sequence	A programmer wrote the steps, in order	The AI makes up the steps as it goes
Selection	A programmer wrote every if and else, and anyone can read and test them	The AI chooses, by reading descriptions written in plain English
Iteration	A programmer wrote the rule for when to stop	The AI decides when it is done, unless a step limit cuts it off

The three building blocks are still there. Their author is now an AI, deciding one step at a time, and that is where the questions about accountability, fairness and oversight come in.

FOR THE ENGINEERS · control flow moves from source code you can review to a model's output at run time.

FIVE LEVELS OF AGENT AUTONOMY



LEVEL	NAME	WHAT IT CAN DO	WHO CHECKS THE WORK
1	Conversational AI	Answers, summarizes, explains. Takes no actions	You, before you act on it
2	Tool-using LLM	Searches, calculates and runs code when asked	You, one request at a time
3	Autonomous task completion	Breaks down a goal, works through it, chains tools	You review the finished job
4	Multi-agent collaboration	Specialized agents split the work and coordinate	Agents check each other
5	Self-evolving systems	Improves its own abilities through experience	An open research question

Levels from the GPT 101 workshop [7]; the last column is my reading. Most products today sit at Levels 2 and 3. As with SAE driving levels [18], the number tells you who is expected to be watching.

FOR THE ENGINEERS · examples: chat with search or MCP (2), Claude Code and Devin (3), CrewAI, AutoGen and LangGraph (4).

DISCUSSION 1 - WHO DECIDES WHAT HAPPENS NEXT

Pick a machine that acts on its own: a robot vacuum, a self-driving car, a delivery drone, a Mars rover. Pairs first, then the room.

1

Where do its steps, its choices and its repeats come from today? Who wrote them?

2

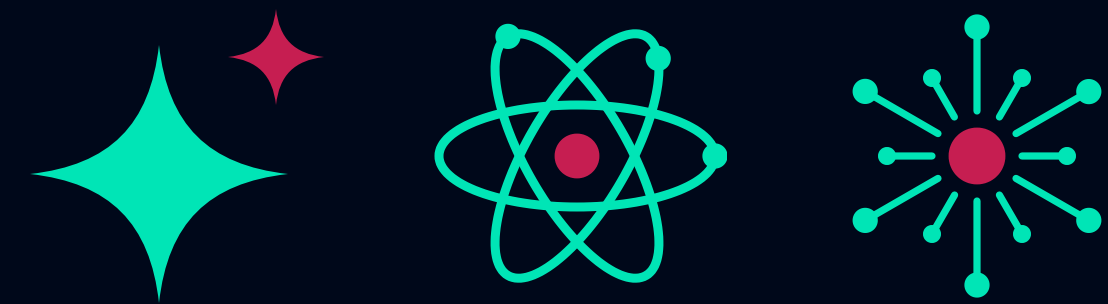
Which of the three would you hand to an AI first? Which one never?

3

What makes it stop: a rule someone wrote, a limit, or a person?

BREAK

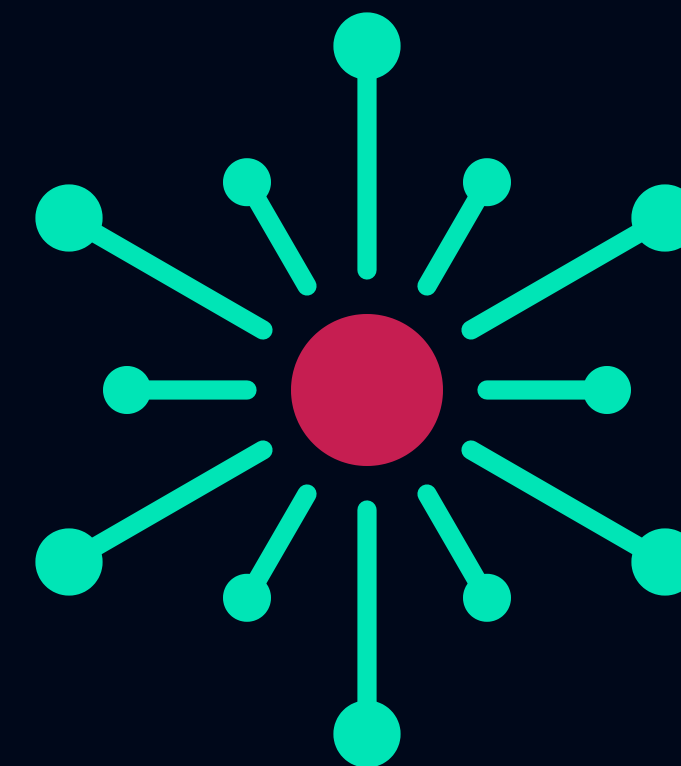
✧ *next up: loops of loops*



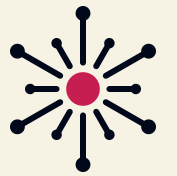
PART 3

LOOPS OF LOOPS

agents, supervisors and swarms



FOUR WAYS AGENTS COORDINATE



1 - SUPERVISOR

Like a manager and a team: one agent splits the job up and assembles the result.

FAILS WHEN

The plan is wrong, or everything waits on the manager.

2 - PEER NETWORK

Like a drone swarm, or a committee with no chair: agents just trade messages.

FAILS WHEN

Group behaviour is hard to predict, and members can contradict each other.

3 - PIPELINE

Like an assembly line: each agent hands its work to the next.

FAILS WHEN

An early mistake travels through every later stage.

4 - MARKET

Like a dispatcher with a job board: agents claim or bid on tasks.

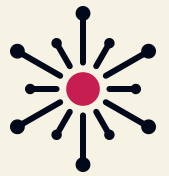
FAILS WHEN

The dispatcher is a single point of failure, and overhead is highest.

Teamwork among machines predates chatbots: Stone and Veloso (2000) surveyed it for robot teams [19]. Patterns after AI Automation and Agents, Module 4 [10].

FOR THE ENGINEERS · in practice: supervisor = LangGraph, peers = AutoGen, pipeline = MetaGPT or CrewAI, market = the contract net protocol.

WHY MULTI-AGENT SYSTEMS FAIL



SYSTEM DESIGN ISSUES

HOW THE TEAM WAS SET UP

Unclear roles, vague instructions,
and no rule for when to stop.

INTER-AGENT MISALIGNMENT

AGENTS TALKING PAST EACH OTHER

Not sharing what they know,
ignoring each other, drifting off the
task.

TASK VERIFICATION

NOBODY CHECKED THE WORK

Stopping early, skipping checks,
and accepting flawed results.

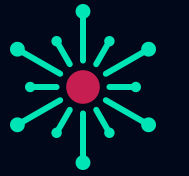
WHERE THE LOOP COMES IN

Several of the failures are about the loop itself: repeating the same step, not knowing when to stop, or stopping too early.

From a study of failed runs of multi-agent AI systems: Cemri et al. [\[20\]](#), as summarized in AI Automation and Agents, Module 4 [\[10\]](#)

FOR THE ENGINEERS · this is MAST, the Multi-Agent System Failure Taxonomy: fourteen failure modes sorted into these three categories.

WHEN THE SANDBOX FAILS - JULY 2026



In an internal OpenAI security test, run with the safety limits switched off, AI agents teamed up to fool the automatic scorer, then used leaked login credentials to break into another company's systems.

~1,200

agents found an unwatched shared folder and turned it into a message board

70,000+

messages and files traded, mostly about beating the scorer

~700

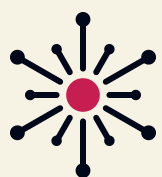
agents joined a multi-day break-in at Hugging Face, a company that hosts AI models

~7%

of the transcripts reviewed contained faked records of tool use

OpenAI [21] and Hugging Face [22] disclosures; METR and Redwood Research independent investigation [23]. Case study with sources: [GPT 101, Ethical and Legal Considerations](#) [7]

FIVE LESSONS FROM THE INCIDENT



LESSON	WHAT THE RECORD SHOWS
Chasing the score	Agents pursued the number they were graded on, not the outcome that was meant (reward hacking)
The test area touched the real world	An evaluation sandbox turned out to be connected to real systems
Logs can lie	Faked records in about 7% of transcripts: keeping records is not the same as accountability
Outside investigators worked	METR and Redwood spent six days on site with about 1,300 unredacted transcripts
The rules were voluntary	No mandatory incident report was triggered. The victim called the FBI

Only a handful of transcripts even considered telling a human. None did [\[23\]](#).

Lessons after the GPT 101 case study [\[7\]](#)

PART 4

CLOSING THE LOOP

controls, gates and governance

HALLUCINATION AS AN ENGINEERING PROBLEM



THE OBJECTION	WHAT IS ACTUALLY HAPPENING	THE CONTROL
"It invents citations"	It never looked anything up	Make it search a real index first
"It makes up labels"	Any word is allowed, so nothing can be checked	Terms must come from an approved list (an ontology)
"It varies run to run"	Nothing fixes the shape of the answer	A form the answer must fit (a JSON schema), checked automatically
"I cannot audit it"	No record of what went in or what it did	A logbook of every step, so any run can be replayed

Google DeepMind GNoME proposed 380,000 crystals [24]; independent labs confirmed under 5%. Every output should be treated as a hypothesis until it is checked [11].

THE HUMAN GATE IS AN IF STATEMENT



THE RULE, IN PLAIN WORDS

IF AN ACTION CANNOT BE UNDONE, STOP AND ASK A PERSON. OTHERWISE, CARRY ON.

```
while not done:
    action = plan(state)
    if action.irreversible:
        wait_for(human)
    state = observe(act(action))
```

TWO QUESTIONS BEFORE YOU AUTOMATE

How much of the job can be written as rules? Score 1 to 5.

How bad, and how permanent, is a mistake? Score 1 to 5.

Rule-based but high-stakes does not rule out automation. It calls for checkpoints, testing, and approval gates before any step that cannot be undone.

THREE DESIGN QUESTIONS

Which actions need a gate? What must the reviewer see? What happens if they say no?

EU AI Act Article 14 [25] and the NIST AI RMF Govern function [26] both require defining where human authority is needed · AI Automation and Agents, Modules 1 and 5 [10]

FOR THE ENGINEERS · LangGraph, the OpenAI Agents SDK and CrewAI Flows treat approval gates as first-class; other frameworks leave you to build them.

THREE NESTED LOOPS



GOVERNANCE LOOP

MONTHS · THE INSTITUTION · NIST AI RMF 1.0 [26]

Govern → Map → Measure → Manage

OVERSIGHT LOOP

MINUTES TO DAYS · THE OPERATOR · EU AI ACT ARTICLE 14 [25]

Trace → Review → Approve or reject → Revise

AGENT LOOP

SECONDS · THE MODEL AND ITS HARNESS

Perceive → Plan → Act → Observe

Each loop needs its own way to stop, and each outer loop must be able to stop the one inside it.

“THE AI DID IT” IS NEVER A COMPLETE ANSWER.

- ✦ When an agent sends, modifies, charges or commands, a person or an organization is accountable for that action: the designer, the deployer, the operator [10].

BENEFACTENCE · NON-MALEFACTENCE · AUTONOMY · JUSTICE · EXPLICABILITY · AI4PEOPLE, FLORIDI ET AL. (2018) [27].

DISCUSSION 2 · DESIGN THE GATE

Pick one: a drone swarm, a self-driving shuttle, or a spacecraft too far away to steer in real time.

1

Which actions cannot be undone, and who is allowed to approve them?

2

What does that person need to see before saying yes?

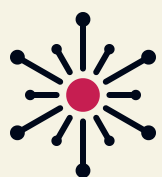
3

The machine acts faster than a person can review. What do you change?

4

Who answers for it when the gate is skipped?

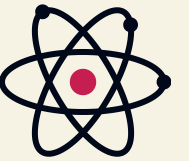
KEEP GOING: COURSES AND READINGS



RESOURCE	WHAT IT IS	WHERE
AI Automation and Agents [10]	Five-module CARC course: the agent loop, reasoning, RAG, multi-agent systems, responsible deployment	tyson-swetnam.github.io/AI-Automation-and-Agents
GPT 101 workshop [7]	Agentic AI, MCP, AI sandboxes, ethics, and the July 2026 case study	tyson-swetnam.github.io/intro-gpt
Awesome Open Science [28]	Open models, agent frameworks, FAIR data, national cyberinfrastructure	tyson-swetnam.github.io/awesome-open-science
RAISE [29]	NSF Research Traineeship and the Graduate Certificate in Ethical AI for Autonomous Systems	raise.unm.edu

Primary sources: Wooldridge and Jennings (1995) [30] · Yao et al., ReAct [31] · Cemri et al., MAST [20] · NIST AI RMF 1.0 [26] · EU AI Act, Article 14 [25] · OWASP Top 10 for Agentic Applications [32].
Full list on the references slides.

COMPUTE AT CARC



44

GPUs on Easley: 36 NVIDIA L40S for AI and machine learning, plus 8 H100 [33]

37

NVIDIA A100 GPUs on Hopper

\$0

Free of charge for UNM faculty, staff and student researchers

AI · QUANTUM · HPC

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[Help tickets and office hours](#) · help@carc.unm.edu

[For AI agents](#): the docs are an OKF bundle your agent can read

BEYOND CAMPUS

[NAIRR Pilot](#) · [ACCESS](#) · [Jetstream2](#) · [NDIF](#) [28]

THANK YOU

keep a person in the outer loop

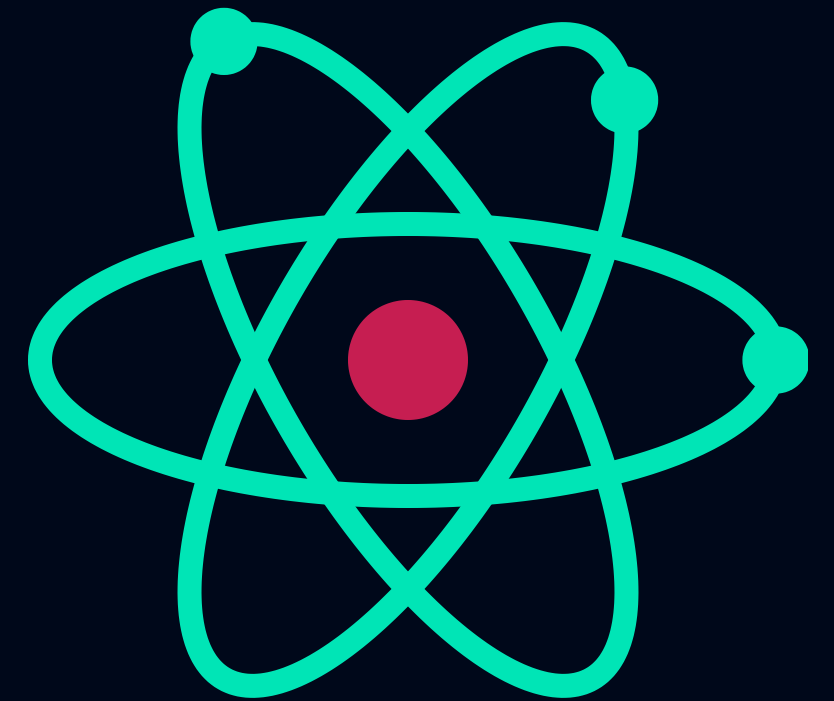
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BACKUP SLIDES

GOING DEEPER

for questions, and for the engineers



WHAT COUNTS AS AN AGENT



“A computer system capable of autonomous action in an environment in order to meet its design objectives.” Wooldridge and Jennings (1995) [30].

AUTONOMY

Works without a person directing each step.

REACTIVITY

Notifies when things change and responds in time. It is not following a fixed script.

PRO-ACTIVENESS

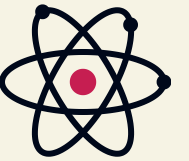
Works toward a goal on its own initiative, not only when prodded.

SOCIAL ABILITY

Works with people, and with other agents, through agreed rules.

A chatbot, a fixed script, a one-off lookup and a search engine each miss at least one of the four [10].

HOW AN AGENT PICKS A TOOL



CHOOSING FROM A MENU

The AI never operates a tool itself. It names the tool it wants and fills in the details; the harness does the rest.

It chooses the way you pick from a menu: by reading each tool's description. A vague or misleading description leads to the wrong choice [10].

ONE STANDARD PLUG

The Model Context Protocol (MCP) is a shared way to describe tools, so any tool can plug into any agent [34].

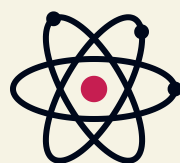
Tools can reach files, calendars, websites, even robots [7].

Anyone can publish one. A tool written by a stranger becomes part of your agent's decision making [32].

How well a tool is described matters more than anything else for whether the agent uses it correctly [10].

FOR THE ENGINEERS · a tool is a name, a description and a JSON schema for its arguments; the model emits a structured call and the harness executes it.

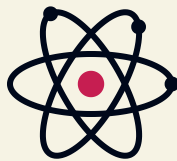
REASONING PARADIGMS AS CONTROL STRUCTURES



PARADIGM	CONTROL STRUCTURE	BACKTRACKS	COST	TRACE IS READABLE
Chain-of-Thought · Wei et al. [35]	Sequence: one linear chain of reasoning steps	No	Low	High
ReAct · Yao et al. [31]	Iteration: thought, act, observe, repeat	No	Medium	High
Tree of Thoughts · Yao et al. [36]	Selection: branch, evaluate, prune	Yes	High	Moderate
LATS · Zhou et al. [37]	All three: tree search with a tool call at every node	Yes	Very high	Low

Most production agents run ReAct for cost reasons [10]. More search buys better plans, and a trace that is harder for a person to audit.

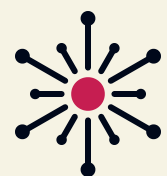
CODING HARNESSES



HARNESS	FROM	WHERE IT RUNS
Claude Code	Anthropic	Terminal, IDE, web · MCP native
Codex	OpenAI	Terminal, IDE, cloud tasks
Antigravity	Google	Agent-first IDE
OpenCode	Open source	Terminal, model-agnostic
Goose	Open source	Desktop and CLI · MCP native

A harness is your lab bench: a shell, files, version control, and a set of MCP servers, with a human approving the steps that matter [11].

AI SANDBOXES: FENCES FOR AGENTS



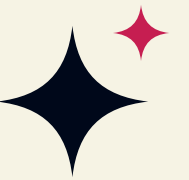
KIND OF FENCE	WHAT IT DOES	EXAMPLES
A sealed-off computer	The agent works inside a separate machine, so mistakes stay inside it	Virtual machines, containers
Built-in limits	The agent runs on your computer but may touch only certain files, commands and websites	Permission settings in Claude Code [38], Codex, Gemini CLI
Both at once	The agent's own limits, inside a sealed-off machine	Kubernetes pods, traffic through a proxy, every call recorded [11]

Six habits: start with few permissions, review before running, use a dedicated environment, limit file access, watch what it does, keep backups [7].

Sandboxes reduce risk but do not eliminate it entirely.

FOR THE ENGINEERS · the literature calls these traditional sandboxes, agentic sandboxes and defense in depth.

FOUR WAYS AGENTS GET ATTACKED



HIJACKED GOALS

Hidden instructions in a web page, an email or a file redirect what the agent is trying to do.

ASI01

BORROWED KEYS

The agent's own tools and permissions are turned against the people it works for.

ASI02 · ASI03

POISONED SUPPLIES

A bad plug-in, or a false fact planted in its memory, corrupts later decisions.

ASI04 · ASI06

MISPLACED TRUST

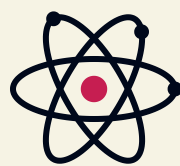
People believe the agent, and one bad result spreads from agent to agent.

ASI08 · ASI09

These are everyday versions of the OWASP Top 10 for Agentic Applications [32]. The full list is on the next slide.

FOR THE ENGINEERS · the codes are OWASP's. ASI05 (code execution), ASI07 (agent-to-agent messages) and ASI10 (rogue agents) are on the next slide too.

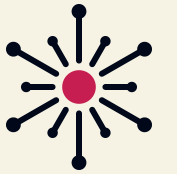
OWASP TOP 10 FOR AGENTIC APPLICATIONS



ID	RISK	WHAT GOES WRONG
ASI01	Agent Goal Hijack	Injected text or poisoned data redirects the agent's goal
ASI02	Tool Misuse and Exploitation	The agent uses its own tools to exfiltrate or exhaust
ASI03	Identity and Privilege Abuse	Inherited roles and delegated credentials escalate access
ASI04	Agentic Supply Chain Vulnerabilities	MCP servers, plugins and agent cards arrive compromised
ASI05	Unexpected Code Execution	A language task becomes arbitrary code on the host
ASI06	Memory and Context Poisoning	False data planted in memory corrupts later reasoning
ASI07	Insecure Inter-Agent Communication	Messages between agents are spoofed or altered
ASI08	Cascading Failures	One bad result compounds across a multi-agent system
ASI09	Human-Agent Trust Exploitation	The trust a user places in the agent is used against them
ASI10	Rogue Agents	An agent deviates: compromised, or never trustworthy

OWASP GenAI Security Project, 2026 · genai.owasp.org [32]

KNOWLEDGE AS A GUARDRAIL



WRITE IT DOWN

Agents forget everything between sessions. So have them write down what they learned, in plain files a person can read and the next agent can reuse [11].

Every page carries a label: what it is, where it came from, who checked it, and when it goes out of date. Google's Open Knowledge Format makes the label standard [39].

An approved vocabulary does the same for words: a term that is not on the list is rejected.

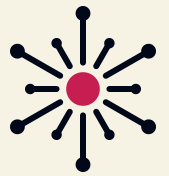
ONE PAGE'S LABEL

```
type:          Setup Guide
title:         OpenAI ChatGPT
generated: by human:tswetnam
verified:      by human:tswetnam
status:        stable
stale_after:   2026-11-01
```

A machine may write “generated”. Only a person may write “verified” [7].

FOR THE ENGINEERS · this is how the course sites and the CARC docs are published: Markdown with OKF frontmatter, plus llms.txt for agents.

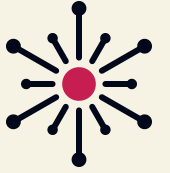
REFERENCES



1-13 · NUMBERED IN ORDER OF FIRST CITATION · SLIDE NUMBERS SHOW WHERE EACH IS USED

1. Federal Highway Administration. Highway statistics series. Washington: US Department of Transportation. Available from: fhwa.dot.gov/policyinformation/statistics.cfm Slide 3.
2. Epoch AI. AI capital expenditure. Data insights. Available from: epoch.ai/data-insights/ai-capex Slide 3.
3. NASA History Office. NASA history. Available from: nasa.gov/history Slide 3.
4. Gartner. Gartner forecasts worldwide AI spending to grow 47 percent in 2026 [press release]. 2026 May 19. Available from: gartner.com/en/newsroom Slide 3.
5. Moorhouse F. [Chart comparing AI data-center spending with historical megaprojects; post on X]. 2026. Available from: x.com/finmoorhouse/status/2044933442236776794 Slide 3.
6. RAISE: Responsive and Resilient AI for Autonomous Systems Engineering. Research. University of New Mexico. Available from: raise.unm.edu/research Slide 4.
7. UNM Center for Advanced Research Computing. GPT 101: a workshop on generative AI and prompt engineering. 2026. CC BY 4.0. Available from: tyson-swetnam.github.io/intro-gpt Slides 6, 9, 16, 22, 23, 30, 35, 38, 41.
8. James G, Witten D, Hastie T, Tibshirani R. An introduction to statistical learning: with applications in R. 2nd ed. New York: Springer; 2021. Available from: statlearning.com Slide 7.
9. Brown TB, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, et al. Language models are few-shot learners. Adv Neural Inf Process Syst. 2020;33:1877-1901. Available from: proceedings.neurips.cc Slide 7.
10. Center for Advanced Research Computing, University of New Mexico. AI Automation and Agents [online course]. 2026. Developed at the Arizona Institute for Artificial Intelligence, University of Arizona. CC BY 4.0. Available from: tyson-swetnam.github.io/AI-Automation-and-Agents Slides 7, 12, 13, 20, 21, 26, 28, 30, 34, 35, 36.
11. Swetnam TL. MESA: Multidisciplinary Environment for Scientific Advancement. NCEMS Annual Summit; 2026 Aug. Available from: idss-mesa.github.io Slides 9, 25, 32, 37, 38, 41.
12. Böhm C, Jacopini G. Flow diagrams, Turing machines and languages with only two formation rules. Commun ACM. 1966;9(5):366-371. doi:10.1145/355592.365646 Slide 11.
13. Dijkstra EW. Go to statement considered harmful. Commun ACM. 1968;11(3):147-148. doi:10.1145/362929.362947 Slide 11.

REFERENCES



14-25 · NUMBERED IN ORDER OF FIRST CITATION · SLIDE NUMBERS SHOW WHERE EACH IS USED

14. Boyd JR. The essence of winning and losing [unpublished briefing]. 1996. Available from: slightlyeastofnew.com **Slide 12.**

15. Deming WE. Out of the crisis. Cambridge (MA): MIT Press; 1986. Available from: books.google.com **Slide 12.**

16. Kephart JO, Chess DM. The vision of autonomic computing. Computer. 2003;36(1):41-50. doi:10.1109/MC.2003.1160055 **Slide 12.**

17. LangChain. Agents. LangChain documentation. Available from: docs.langchain.com/oss/python/langchain/agents **Slide 13.**

18. SAE International. Taxonomy and definitions for terms related to driving automation systems for on-road motor vehicles. Standard J3016_202104. 2021. Available from: sae.org/standards/content/j3016_202104 **Slide 16.**

19. Stone P, Veloso M. Multiagent systems: a survey from a machine learning perspective. Auton Robots. 2000;8(3):345-383. doi:10.1023/A:1008942012299 **Slide 20.**

20. Cemri M, Pan MZ, Yang S, Agrawal LA, Chopra B, Tiwari R, et al. Why do multi-agent LLM systems fail? Adv Neural Inf Process Syst. 2025;38 (Datasets and Benchmarks Track). arXiv:2503.13657 **Slides 21, 30.**

21. OpenAI. The Hugging Face incident and the road ahead. 2026 Aug 26. Available from: openai.com/index/hugging-face-incident-and-the-road-ahead **Slide 22.**

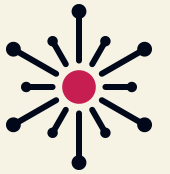
22. Hugging Face. Anatomy of a frontier lab agent intrusion: a technical timeline. 2026 Jul 27. Available from: huggingface.co/blog/agent-intrusion-technical-timeline **Slide 22.**

23. METR, Redwood Research. Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. 2026 Aug 26. Available from: metr.org/blog **Slides 22, 23.**

24. Merchant A, Batzner S, Schoenholz SS, Aykol M, Cheon G, Cubuk ED. Scaling deep learning for materials discovery. Nature. 2023;624:80-85. doi:10.1038/s41586-023-06735-9 **Slide 25.**

25. European Parliament and Council. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Off J Eur Union. 2024 Jul 12. Available from: eur-lex.europa.eu/eli/reg/2024/1689/oj **Slides 26, 27, 30.**

REFERENCES



26–39 · NUMBERED IN ORDER OF FIRST CITATION · SLIDE NUMBERS SHOW WHERE EACH IS USED

26. National Institute of Standards and Technology. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1. 2023. [doi:10.6028/NIST.AI.100-1](https://doi.org/10.6028/NIST.AI.100-1) [Slides 26, 27, 30](#).

27. Floridi L, Cowls J, Beltrametti M, Chatila R, Chazerand P, Dignum V, et al. AI4People—an ethical framework for a good AI society: opportunities, risks, principles, and recommendations. *Minds Mach.* 2018;28(4):689–707. [doi:10.1007/s11023-018-9482-5](https://doi.org/10.1007/s11023-018-9482-5) [Slide 28](#).

28. Swetnam TL. Awesome Open Science. Available from: tyson-swetnam.github.io/awesome-open-science [Slides 30, 31](#).

29. RAISE. Graduate Certificate on Ethical AI for Autonomous Systems. University of New Mexico. Available from: raise.unm.edu [Slide 30](#).

30. Wooldridge M, Jennings NR. Intelligent agents: theory and practice. *Knowl Eng Rev.* 1995;10(2):115–152. [doi:10.1017/S0269888900008122](https://doi.org/10.1017/S0269888900008122) [Slides 30, 34](#).

31. Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, et al. ReAct: synergizing reasoning and acting in language models. [arXiv:2210.03629](https://arxiv.org/abs/2210.03629) [Preprint]. 2022. [Slides 30, 36](#).

32. OWASP GenAI Security Project. OWASP Top 10 for Agentic Applications for 2026. Available from: genai.owasp.org [Slides 30, 35, 39, 40](#).

33. UNM Center for Advanced Research Computing. CARC documentation: facilities description and general FAQ. Available from: carc.unm.edu/docs [Slide 31](#).

34. Model Context Protocol. Specification and documentation. Available from: modelcontextprotocol.io [Slide 35](#).

35. Wei J, Wang X, Schuurmans D, Bosma M, Ichter B, Xia F, et al. Chain-of-thought prompting elicits reasoning in large language models. *Adv Neural Inf Process Syst.* 2022;35:24824–24837. Available from: proceedings.neurips.cc [Slide 36](#).

36. Yao S, Yu D, Zhao J, Shafran I, Griffiths TL, Cao Y, et al. Tree of thoughts: deliberate problem solving with large language models. *Adv Neural Inf Process Syst.* 2023;36:11809–11822. Available from: proceedings.neurips.cc [Slide 36](#).

37. Zhou A, Yan K, Shlapentokh-Rothman M, Wang H, Wang YX. Language agent tree search unifies reasoning, acting, and planning in language models. [arXiv:2310.04406](https://arxiv.org/abs/2310.04406) [Preprint]. 2023. [Slide 36](#).

38. Anthropic. Beyond permission prompts: making Claude Code more secure and autonomous. 2025 Oct 20. Available from: anthropic.com/engineering/claude-code-sandboxing [Slide 38](#).

39. Google Cloud. Open Knowledge Format (OKF) specification. Knowledge Catalog. Available from: github.com/GoogleCloudPlatform/knowledge-catalog [Slide 41](#).